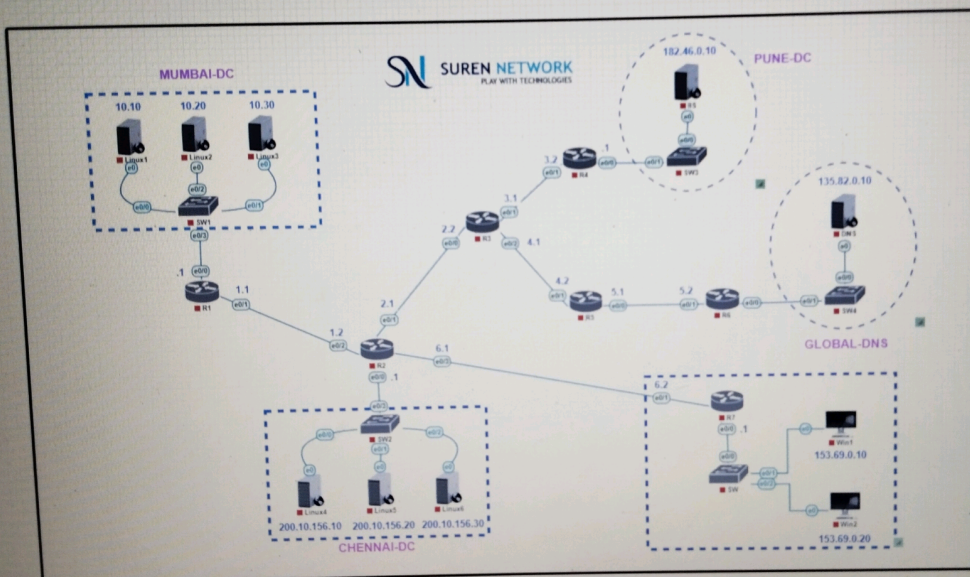




R1

```
R1(config)#ip route 5.0.0.0 255.0.0.0 1.0.0.2
R1(config)#ip route 6.0.0.0 255.0.0.0 1.0.0.2
R1(config)#ip route 135.82.0.0 255.255.0.0 1.0.0.2
R1(config)#ip route 153.69.0.0 255.255.0.0 1.0.0.2
R1(config)#ip route 182.46.0.0 255.255.0.0 1.0.0.2
R1(config)#ip route 200.10.156.0 255.255.255.0 1.0.0.2
R1(config)#exit
```

R1#

*Aug 30 04:53:28.780: %SYS-5-CONFIG_I: Configured from console by console

R1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    1.0.0.0/8 is directly connected, Ethernet0/1
L    1.0.0.1/32 is directly connected, Ethernet0/1
S    2.0.0.0/8 [1/0] via 1.0.0.2
S    3.0.0.0/8 [1/0] via 1.0.0.2
S    4.0.0.0/8 [1/0] via 1.0.0.2
S    5.0.0.0/8 [1/0] via 1.0.0.2
S    6.0.0.0/8 [1/0] via 1.0.0.2
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Ethernet0/0
L    10.0.0.1/32 is directly connected, Ethernet0/0
S    135.82.0.0/16 [1/0] via 1.0.0.2
S    153.69.0.0/16 [1/0] via 1.0.0.2
S    182.46.0.0/16 [1/0] via 1.0.0.2
S    200.10.156.0/24 [1/0] via 1.0.0.2
```

R1#

R1#

R1#

R1#

R1#

R1#sh ip int br

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	10.0.0.1	YES	manual	up	up
Ethernet0/1	1.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	up	up
Ethernet0/3	unassigned	YES	TFTP	up	up

R1#

R2

```
R2(config)#ip route 153.69.0.0 255.255.0.0 6.0.0.2
R2(config)#ip route 182.46.0.0 255.255.0.0 2.0.0.2
R2(config)#exit
R2#
```

*Aug 30 05:06:33.382: %SYS-5-CONFIG_I: Configured from console by console

```
R2#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
o001					
Ethernet0/0	200.10.156.1	YES	manual	up	up
Ethernet0/1	2.0.0.1	YES	manual	up	up
Ethernet0/2	1.0.0.2	YES	manual	up	up
Ethernet0/3	6.0.0.1	YES	manual	up	up

```
R2#sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    1.0.0.0/8 is directly connected, Ethernet0/2
L    1.0.0.2/32 is directly connected, Ethernet0/2
2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    2.0.0.0/8 is directly connected, Ethernet0/1
L    2.0.0.1/32 is directly connected, Ethernet0/1
S    3.0.0.0/8 [1/0] via 2.0.0.2
S    4.0.0.0/8 [1/0] via 2.0.0.2
S    5.0.0.0/8 [1/0] via 2.0.0.2
6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    6.0.0.0/8 is directly connected, Ethernet0/3
L    6.0.0.1/32 is directly connected, Ethernet0/3
S    10.0.0.0/8 [1/0] via 1.0.0.1
S    135.82.0.0/16 [1/0] via 2.0.0.2
S    153.69.0.0/16 [1/0] via 6.0.0.2
S    182.46.0.0/16 [1/0] via 2.0.0.2
200.10.156.0/24 is variably subnetted, 2 subnets, 2 masks
C    200.10.156.0/24 is directly connected, Ethernet0/0
L    200.10.156.1/32 is directly connected, Ethernet0/0
R2#
```

R3

```
R3(config)#ip route 200.10.156.0 255.255.255.0 2.0.0.1
```

```
R3(config)#
```

```
R3(config)#exit
```

```
R3#sh
```

```
*Aug 30 05:25:05.504: %SYS-5-CONFIG_I: Configured from console by console
```

```
R3#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
oc0/1					
Ethernet0/0	2.0.0.2	YES	manual	up	up
Ethernet0/1	3.0.0.1	YES	manual	up	up
Ethernet0/2	4.0.0.1	YES	manual	up	up
Ethernet0/3	unassigned	YES	TFTP	up	up

```
R3#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2
```

```
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
```

```
ia - IS-IS inter area, * - candidate default, U - per-user static route
```

```
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
```

```
a - application route
```

```
+ - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S    1.0.0.0/8 [1/0] via 2.0.0.1
    2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    2.0.0.0/8 is directly connected, Ethernet0/0
L    2.0.0.2/32 is directly connected, Ethernet0/0
    3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    3.0.0.0/8 is directly connected, Ethernet0/1
L    3.0.0.1/32 is directly connected, Ethernet0/1
    4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    4.0.0.0/8 is directly connected, Ethernet0/2
L    4.0.0.1/32 is directly connected, Ethernet0/2
S    6.0.0.0/8 [1/0] via 2.0.0.1
S    10.0.0.0/8 [1/0] via 2.0.0.1
S    135.82.0.0/16 [1/0] via 4.0.0.2
S    153.69.0.0/16 [1/0] via 2.0.0.1
    182.46.0.0/24 is subnetted, 1 subnets
S    182.46.0.0 [1/0] via 3.0.0.2
S    200.10.156.0/24 [1/0] via 2.0.0.1
```

```
R3#
```


R4

% Ambiguous command: "ex"

R4(config)#no shutdown

% Invalid input detected at '^' marker.

R4(config)#exit

R4#

'Aug 30 05:38:29.527: %SYS-5-CONFIG_I: Configured from console by console

R4#sh ip int br

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	3.0.0.2	YES	manual	up	up
Ethernet0/1	182.46.0.10	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	up	up
Ethernet0/3	unassigned	YES	TFTP	up	up

R4#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 3.0.0.1
S    2.0.0.0/8 [1/0] via 3.0.0.1
      3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      3.0.0.0/8 is directly connected, Ethernet0/0
L      3.0.0.2/32 is directly connected, Ethernet0/0
S    4.0.0.0/8 [1/0] via 3.0.0.1
S    5.0.0.0/8 [1/0] via 3.0.0.1
S    6.0.0.0/8 [1/0] via 3.0.0.1
S   10.0.0.0/8 [1/0] via 3.0.0.1
S   135.82.0.0/16 [1/0] via 3.0.0.1
S   153.69.0.0/16 [1/0] via 3.0.0.1
      182.46.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      182.46.0.0/16 is directly connected, Ethernet0/1
L      182.46.0.10/32 is directly connected, Ethernet0/1
S   200.10.156.0/24 [1/0] via 3.0.0.1
```

R4#

R5

```
R5(config)#ip route 182.46.0.0 255.255.0.0 4.0.0.1
```

```
R5(config)#ip route 135.82.0.0 255.255.0.0 5.0.0.2
```

```
R5(config)#ip route 153.69.0.0 255.255.0.0 4.0.0.1
```

```
R5(config)#
```

```
R5(config)#exit
```

```
R5#
```

```
*Aug 30 06:03:45.860: %SYS-5-CONFIG_I: Configured from console by console
```

```
R5#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
ocol					
Ethernet0/0	5.0.0.1	YES	manual	up	up
Ethernet0/1	4.0.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	up	up
Ethernet0/3	unassigned	YES	TFTP	up	up

```
R5#sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

```
Gateway of last resort is not set
```

```
S    1.0.0.0/8 [1/0] via 4.0.0.1
S    2.0.0.0/8 [1/0] via 4.0.0.1
S    3.0.0.0/8 [1/0] via 4.0.0.1
    4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    4.0.0.0/8 is directly connected, Ethernet0/1
L    4.0.0.2/32 is directly connected, Ethernet0/1
    5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    5.0.0.0/8 is directly connected, Ethernet0/0
L    5.0.0.1/32 is directly connected, Ethernet0/0
S   10.0.0.0/8 [1/0] via 4.0.0.1
    18.0.0.0/16 is subnetted, 1 subnets
S    18.46.0.0 [1/0] via 4.0.0.1
S   135.82.0.0/16 [1/0] via 5.0.0.2
S   153.69.0.0/16 [1/0] via 4.0.0.1
S   182.46.0.0/16 [1/0] via 4.0.0.1
S   200.10.156.0/24 [1/0] via 4.0.0.1
```

```
R5#
```


R7

```
R7(config)#ip route 4.0.0.0 255.0.0.0 6.0.0.1
R7(config)#ip route 200.10.156.0 255.255.255.0 6.0.0.1
R7(config)#ip route 5.0.0.0 255.0.0.0 6.0.0.1
R7(config)#ip route 182.46.0.0 255.255.0.0 6.0.0.1
R7(config)#ip route 135.82.0.0 255.255.0.0 6.0.0.1
R7(config)#exit
R7#
```

*Aug 30 06:14:46.540: %SYS-5-CONFIG_I: Configured from console by console

```
R7#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
ocol					
Ethernet0/0	153.69.0.1	YES	manual	up	up
Ethernet0/1	6.0.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	up	up
Ethernet0/3	unassigned	YES	TFTP	up	up

```
R7#sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    1.0.0.0/8 [1/0] via 6.0.0.1
S    2.0.0.0/8 [1/0] via 6.0.0.1
S    3.0.0.0/8 [1/0] via 6.0.0.1
S    4.0.0.0/8 [1/0] via 6.0.0.1
S    5.0.0.0/8 [1/0] via 6.0.0.1
    6.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      6.0.0.0/8 is directly connected, Ethernet0/1
L      6.0.0.2/32 is directly connected, Ethernet0/1
S    10.0.0.0/8 [1/0] via 6.0.0.1
S    135.82.0.0/16 [1/0] via 6.0.0.1
    153.69.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      153.69.0.0/16 is directly connected, Ethernet0/0
L      153.69.0.1/32 is directly connected, Ethernet0/0
S    182.46.0.0/16 [1/0] via 6.0.0.1
S    200.10.156.0/24 [1/0] via 6.0.0.1
```

```
R7#
```

```
Router(config)#ip route 135.82.0.0 255.255.0.0 5.0.0.2
```

```
% Incomplete command.
```

```
Router(config)#ip route 135.82.0.0 255.255.0.0 5.0.0.2
```

```
Router(config)#ip route 153.69.0.0 255.255.0.0 5.0.0.2
```

```
Router(config)#exit
```

```
Router#
```

```
*Aug 30 06:29:09.899: %SYS-5-CONFIG_I: Configured from console by console
```

```
Router#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
ooc1					
Ethernet0/0	5.0.0.1	YES	manual	up	up
Ethernet0/1	4.0.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	unset	administratively down	down
Ethernet0/3	unassigned	YES	unset	administratively down	down

```
Router#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr
```

```
Gateway of last resort is not set
```

```
S    1.0.0.0/8 [1/0] via 5.0.0.2
S    2.0.0.0/8 [1/0] via 5.0.0.2
S    3.0.0.0/8 [1/0] via 5.0.0.2
S    4.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      4.0.0.0/8 is directly connected, Ethernet0/1
L      4.0.0.2/32 is directly connected, Ethernet0/1
S    5.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      5.0.0.0/8 is directly connected, Ethernet0/0
L      5.0.0.1/32 is directly connected, Ethernet0/0
S    6.0.0.0/8 [1/0] via 5.0.0.2
S   10.0.0.0/8 [1/0] via 5.0.0.2
S   135.82.0.0/16 [1/0] via 5.0.0.2
S   153.69.0.0/16 [1/0] via 5.0.0.2
S   182.46.0.0/16 [1/0] via 5.0.0.2
S   200.10.156.0/24 [1/0] via 5.0.0.2
```

```
Router#
```

```
Router#
```



```
QEMU (Linux1)
ls /etc/sysconfig/network-scripts/
vi /etc/sysconfig/network-scripts/ifcfg-eth0

DEVICE=eth0
BOOTPROTO=none
ONBOOT=yes
IPADDR=10.0.0.10
NETMASK=255.0.0.0
GATEWAY=10.0.0.1
BROADCAST=10.255.255.255
DHCP=no

/etc/network/interfaces [Modified] 16/16 100%
```

BAI-DC

PUNE-DC

CHENNAI-DC



C:\WINDOWS\system32\cmd.exe

Ethernet adapter Local Area Connection:

```
Connection-specific DNS Suffix . :  
IP Address. . . . . : 153.69.0.10  
Subnet Mask . . . . . : 255.255.0.0  
Default Gateway . . . . . : 153.69.0.1
```

C:\Documents and Settings\user>ping 1.0.0.1

Pinging 1.0.0.1 with 32 bytes of data:

```
Reply from 1.0.0.1: bytes=32 time=4ms TTL=253  
Reply from 1.0.0.1: bytes=32 time=2ms TTL=253  
Reply from 1.0.0.1: bytes=32 time=2ms TTL=253  
Reply from 1.0.0.1: bytes=32 time=2ms TTL=253
```

Ping statistics for 1.0.0.1:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 2ms, Maximum = 4ms, Average = 2ms
```

C:\Documents and Settings\user>ping 4.0.0.1

Pinging 4.0.0.1 with 32 bytes of data:

```
Reply from 4.0.0.1: bytes=32 time=2ms TTL=253  
Reply from 4.0.0.1: bytes=32 time=1ms TTL=253  
Reply from 4.0.0.1: bytes=32 time=2ms TTL=253  
Reply from 4.0.0.1: bytes=32 time=1ms TTL=253
```

Ping statistics for 4.0.0.1:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

C:\Documents and Settings\user>ping 3.0.0.1

Pinging 3.0.0.1 with 32 bytes of data:

```
Reply from 3.0.0.1: bytes=32 time=1ms TTL=253  
Reply from 3.0.0.1: bytes=32 time=3ms TTL=253  
Reply from 3.0.0.1: bytes=32 time=3ms TTL=253  
Reply from 3.0.0.1: bytes=32 time=2ms TTL=253
```

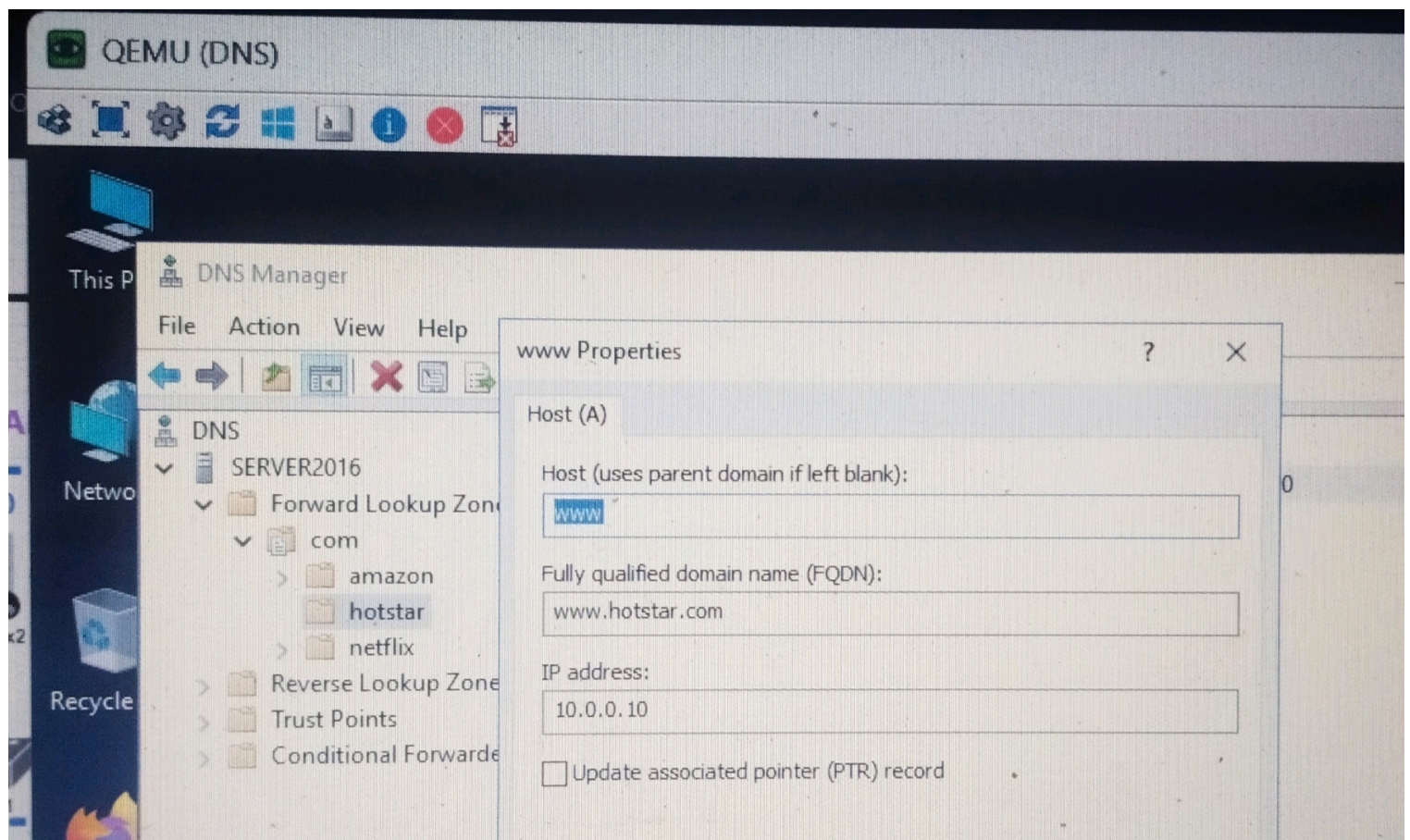
Ping statistics for 3.0.0.1:

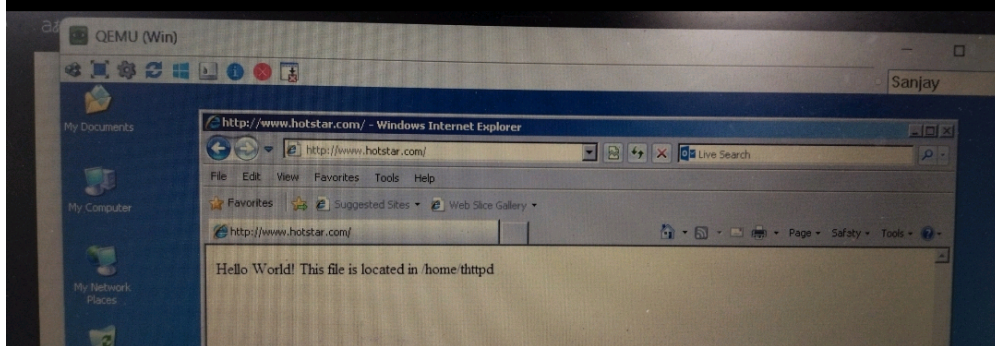
```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 1ms, Maximum = 3ms, Average = 2ms
```

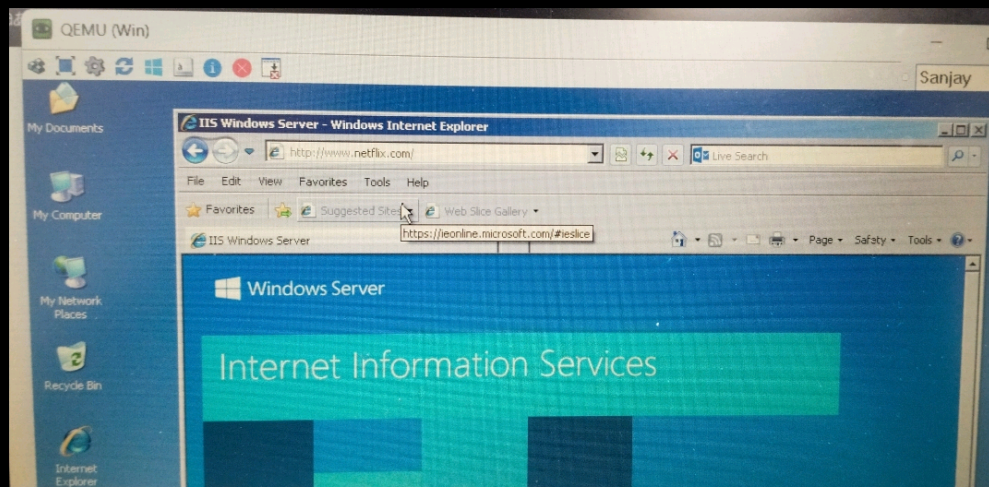
C:\Documents and Settings\user>ping 6.0.0.1

Pinging 6.0.0.1 with 32 bytes of data:

Reply from 6.0.0.1: bytes=32 time=4ms TTL=254







Redirecting to the Mumbai data center's Web server.

